Orientation-Aware Vehicle Detection in Aerial Images via an Anchor-Free Object Detection Approach

Furong Shi, Tong Zhang, and Tao Zhang

***Abstract*— Vehicle detection in aerial images is an important and challenging task in the field of remote sensing. Recently, deep learning technologies have yielded superior performance for object detection in remote sensing images. However, the detection results of the existing methods are horizontal bounding boxes that ignore vehicle orientations, thereby having limited applicability in scenes with dense vehicles or clutter backgrounds. In this article, we propose a one-stage, anchor-free detection approach to detect arbitrarily oriented vehicles in high-resolution aerial images. The vehicle detection task is transformed into a multitask learning problem by directly predicting high-level vehicle features via a fully convolutional network. That is, a classification subtask is created to look for vehicle central points and three regression subtasks are created to predict vehicle orientations, scales, and offsets of vehicle central points. First, coarse and fine feature maps outputted from different stages of a residual network are concatenated together by a feature pyramid fusion strategy. Upon the concatenated features, four convolutional layers are attached in parallel to predict high-level vehicle features. During training, task uncertainty learned from the training data is used to weight loss function in the multitask learning setting. For inferencing, oriented bounding boxes are generated using the predicted vehicle features, and oriented nonmaximum suppression (NMS) postprocessing is used to reduce redundant results. Experiments on two public aerial image data sets have shown the effectiveness of the proposed approach.**

***Index Terms*— Anchor-free, convolutional neural network (CNN), multitask learning, orientation prediction, vehicle detection.**

I. Introduction

With the widespread availability of unmanned aerial vehicles (UAVs), high-resolution aerial images can be easily obtained. Due to their capability to provide full coverage of the traffic situation, vehicle detection in high-resolution aerial images has been attracting significant attention in the field of remote sensing and playing an important role in many applications such as intelligent transportation systems [1], traffic modeling [2], traffic monitoring [3], [4], environment monitoring [5], and urban planning [6]. However, detecting vehicles in high-resolution aerial images is still a challenging task mainly caused by a large number of similar structures of other objects (e.g., air conditioning units on the roof, trash bins, and road marks), complicated backgrounds, and occlusions from the shadows of trees and buildings.

In recent years, deep learning technologies, especially convolutional neural network (CNN), have made great achievement in object detection [7], [8] (e.g., region-based CNN (R-CNN) [11], faster-R-CNN [16], single-shot multibox detector (SSD) [20]). Although the CNN-based object detection methods have achieved great success in natural scenes, there are two disadvantages for applying it to vehicle detection in aerial images. On one hand, the detection results of the existing methods are usually represented by horizontal bounding boxes which ignore vehicle orientations, thereby having limited applicability in scenes with dense vehicles or clutter backgrounds. Meanwhile, the ignored vehicle orientations are also an important vehicle attribute and meaningful for traffic management. For example, when traffic incidents happen, there usually will be one or more vehicles with irregular angles on the road. Accordingly, the vehicle orientation and traffic flow can be served as important features to detect traffic incidents in road traffic monitoring system [9]. Since orientations provide additional information for encoding vehicle features, they are also useful for robust wide-area video object tracking [10]. On the other hand, deep learning vehicle detection approaches are generally based on anchor box prediction. However, the use of anchor boxes introduces many hyper-parameters and design choices such as number of boxes, sizes, and aspect ratios of boxes. These choices need to be carefully designed and become even more complicated in aerial images with multiple resolutions.

In this article, we propose a one-stage, anchor-free approach for orientation-aware vehicle detection, which is composed of four subtasks to directly predict high-level vehicle features, that is, one classification subtask to identify vehicle central points and two regression subtasks to predict vehicle shape (i.e., scale and orientation), and an extra regression subtask to

Manuscript received March 23, 2020; revised May 12, 2020 and June 18, 2020; accepted July 16, 2020. Date of publication August 6, 2020; date of current version May 21, 2021. This work was supported in part by the National Key Research and Development Program of China (International Scientific & Technological Cooperation Program) under Grant 2019YFE0106500 and in part by the National Natural Science Foundation of China under Grant 41871308. *(Corresponding author: Tong Zhang.)*

Furong Shi and Tong Zhang are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: shifurong@whu.edu.cn; zhangt@whu.edu.cn).

Tao Zhang is with the Tianjin Institute of Geotechnical Investigation Surveying, Tianjin 300191, China (e-mail: zhangtao437@163.com).

Color versions of one or more of the figures in this article are available online at https://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.3011418

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.

slightly adjust the vehicle center location. Specifically, coarse and fine features from different stages of a residual network are concatenated together using a feature pyramid fusion strategy, and then the fused features are parsed by four parallel convolutional layers to produce high-level vehicle feature predictions. Inspired by Kendall *et al.* [11], task uncertainty is modeled by the weighting multitask loss and learning from the training data. During inferencing, the oriented bounding boxes can be generated with the predicted vehicle features. To reduce the redundant results, the oriented nonmaximum suppression (NMS) which computes intersection over union (IoU) on the oriented bounding boxes is adapted during postprocessing. We obtain superior performances on two public aerial image data sets (i.e., DLR Munich vehicle data set [24] and vehicle detection in aerial imagery (VEDAI) data set [41]).

The contributions of this article are summarized as follows.

1) An orientation-aware vehicle detector is proposed, which can directly provide vehicle orientation in results. The orientation prediction improves the detection accuracy particularly for scenes with dense vehicles.
2) An anchor-free vehicle detection method is introduced. It does not require a dedicated module for generating proposals (e.g., region proposal network, RPN), thereby avoiding the hyper-parameter tuning for predefined anchor boxes (e.g., number, size, and aspect ratio).
3) A multitask learning procedure is developed to automatically learn the weights of the multitask loss function from the training data according to the uncertainty of each subtask, and thus does not introduce additional hyper-parameters of weight scalars.

The remainder of this article is organized as follows. The related work is briefly reviewed in Section II. The detail of the proposed method is introduced in Section III, and the experimental results are reported and analyzed in Section IV. Finally, a conclusion for this article is presented in Section V.

II. Related Work

In this section, the relevant works of traditional vehicle detection, CNN-based generic object detection, and CNN-based vehicle detection are briefly reviewed.

A. Traditional Manual Vehicle Detection in Aerial Images

Being an important and challenging task in remote sensing, vehicle detection in aerial images has received considerable attention. Traditionally, handcrafted features extracted from sliding windows are used in vehicle detection [29]–[40]. Early methods exploit features that represented vehicle profiles to detect vehicles in relatively low-resolution satellite images. For example, Zhao and Nevatia [29] incorporate shadows and boundaries of vehicles into a Bayesian network to detect vehicles. With the widespread availability of high-resolution aerial images, more accurate vehicle detection methods have been developed. These detectors take vehicle details into account using shape and texture features. Moranduzzo and Melgani [30]–[32] propose several vehicle detection methods using scale-invariant feature transform (SIFT) or histograms of oriented gradients (HOG) features and support vector machine (SVM) classifier. Chen *et al.* [33] make use of HOG feature and sparse representation dictionary to detect vehicles. Local steering kernel (LSK) features are aggregated by Bag-of-Words (BoW) representation, and then SVM classifier is trained to class each vehicle candidate in [34]. Some works in [35]–[37] integrate multiple visual features to detect vehicles. Tan *et al.* [38] and Leitloff *et al.* [39] introduce road areas from the geographic information system (GIS) data as priori knowledge to detect vehicle on road. Liu and Mattyus [40] present an effective vehicle extraction method using an integral channel concept with Haar-like features and an AdaBoost classifier in a soft-cascade structure without any additional information. These traditional vehicle detection methods based on handcrafted features heavily rely on the expert knowledge in feature design. With the increasing challenge for vehicle detection in complicated scenes, the representation capabilities of these shallow features become limited.

B. CNN-Based General Object Detection Methods

Since the R-CNN model was first introduced in 2014, deep CNN-based models have played a central role in generic object detection task. The CNN-based object detection methods can be divided into two-stage detection and one-stage detection [8].

The two-stage detectors generate a sparse set of regions of interest (RoIs) and then classify and refine each of them by deep CNN. R-CNN [12] generates RoIs using selective search algorithm [13]. Spatial pyramid pooling networks (SPPNet) [14] and fast R-CNN [15] improve R-CNN by sharing convolutional layers between object proposals for an image. Faster R-CNN [16] can be trained end-to-end by designing RPN to generate candidate regions and a classifier after RPN to further infer the candidates. Feature pyramid network (FPN) [17] fuses multilevel features in the framework of faster R-CNN. The one-stage detectors remove the RoIs generating step and directly predict bounding boxes and class scores. You only look once (YOLO) [18] divides the input image into grids to simultaneously do regression and classification for each grid. YOLOv2 [19] improves the performance of YOLO using anchor boxes. SSD [20] predicts bounding boxes and class scores from multiple feature scales. Reverse Connection with Objectness Prior Networks (RON) [21] enhances the combination of low-level and high-level features to predict bounding boxes more accurately. RetinaNet [22] addresses the positive and negative anchor boxes imbalance issue of dense detectors using a focal loss function. Vertical Spatial Sequence Attention Network (VSSA-NET) [23] is an effective single-stage detector using the attention mechanism to avoid region proposal process. These methods achieve great performance in general object detection. However, they are based on anchor box predictions, which require tedious configurations in anchors.

Recently, the anchor-free detection methods, which are free from anchor boxes, have attracted a surge of research interest. CornerNet [24] is an anchor-free one-stage detector, which uses a single CNN to detect the top-left corner and the bottom-right corner of an object bounding box and then group them to form final detected results using associative embedding [25]. CenterNet [26] detects an object as a triplet including the center point, the top-left corner, and the bottom-right

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.

corner of the object. Compared with CornerNet, the CenterNet effectively reduces incorrect bounding boxes via exploring central regions. ExtremeNet [27] predicts four extreme points and a center point, and then directly groups the five key points into a tighter object bounding box. CSP [28] detects the center point and the scale of objects, thereby avoiding complex postprocessing based on key point pairs. The anchor-free detection methods are fast and also maintain good performance. However, as vehicles are arbitrarily oriented in aerial images and are different from general objects in natural scenes, the CNN-based general object detectors cannot be directly used in vehicle detection task.

C. CNN-Based Vehicle Detection in Aerial Images

With the application of CNN in generic object detection, deep CNN technology is also widely used in vehicle detection in aerial images. Early CNN-based methods [43]–[45] still use traditional vehicle detection framework, which generates vehicle candidates via sliding windows and then infer each candidate. Chen *et al.* [43] propose a vehicle detection method using sliding-window search technique to generate candidates and a deep CNN to extract features and classify each candidate into vehicles and nonvehicles. Qu *et al.* [44] apply the multiscale SPP to capture multiscale image features for vehicle detection. Ammour *et al.* [45] first use a super pixels segmentation technique to generate vehicle candidate regions, then extract the features by a pretrained deep CNN, and finally classify each candidate with an SVM classifier. These methods improve the performance of vehicle detection using deep features, but cannot be trained end to end.

Later, some end-to-end vehicle detection methods based on faster R-CNN [46]–[48] are proposed, which use anchor boxes to generate candidates. Tang *et al.* [46] and Deng *et al.* [47] develop a vehicle proposal network to generate vehicle candidates and a vehicle classification network to verify them within the framework of faster R-CNN. Tang *et al.* [48] improve the vehicle detection performance by proposing a hyper region proposal network (HRPN) with a combination of hierarchical feature maps to extract vehicle candidates and a cascade of boosted classifier to classify each candidate. Tang *et al.* [49] present an end-to-end single CNN based on SSD to predict vehicle locations and orientations simultaneously. Zheng *et al.* [50] use vehicle synthesis generative adversarial networks (VS-GANs) to argument the training data and greatly improve the performance of vehicle detection. In [51], a recurrent attention structure is proposed to focus on the key regions of a scene such as vehicles and airplanes. These methods archive better performance than traditional detectors. However, they are based on anchor box predictions, with the need to design hyper-parameters of anchor boxes such as number, sizes, and aspect ratios of boxes. Furthermore, these methods generally fail to capture accurate vehicle orientations and have a limited applicability in scenes with dense vehicles. In this work, arbitrarily oriented vehicles in aerial images are detected by a one-stage and anchor-free method.

III. Methodology

The framework of the proposed method, denoted as vehicle center, scale, and orientation prediction-based detector (VCSOP detector), is illustrated in Fig. 1. It mainly includes four modules, namely, convolutional feature extraction and fusion, multitask learning for high-level vehicle feature predictions, an uncertainty weighted loss function for training, and postprocessing of the detection results.

In this work, coarse and fine convolutional features are obtained from different stages of a residual network and combined together through a feature pyramid fusion strategy. Upon the availability of the fused features, a multitask learning module is used to parse the fused features into high-level vehicle feature predictions. During training, the information of four subtasks is incorporated into a single loss function, which is weighted using homoscedastic task uncertainty [11]. The vehicle inference process is introduced including the visualization of the oriented vehicle detection results using the outputs of high-level vehicle feature predictions and a specialized postprocessing method based on NMS to reduce redundant results.

A. Convolutional Feature Extraction and Fusion

Fig. 2 shows the convolutional feature extraction and fusion network, which takes aerial images as inputs and outputs the fused pyramid feature maps at different scales. This network involves a residual network to extract coarse and fine convolutional features and a feature pyramid fusion strategy to combine multiscale feature representation.

The convolutional feature extraction network is based on a residual network of 50 layers (ResNet-50) [52], which divides its convolutional layers into five stages. The outputs of each stage are denoted as C1, C2, C3, C4, and C5. A dilated convolution layer is applied in the last stage to maintain the top feature map as 1/16 of the input size. Further details about the residual network to extract coarse and fine convolutional features ate provided in Appendix A.

It is known that higher layer features contain global semantic information, whereas lower layer features provide precise localization information [53]. To localize vehicles precisely, a feature pyramid fusion strategy based on FPN [16] is used to combine coarse and fine convolutional feature maps that are extracted from different stages of the above residual network. More specifically, the feature maps of C3, C4, and C5 are incorporated. By the feature fusion strategy, a pyramid of feature maps (P3, P4, and P5) is first constructed. Then P3, P4, and P5 are up-sampled to the 1/4 spatial resolution of the input image. Finally, these up-sampled features with the same resolution are fused together using a concatenation operation, and the fused feature maps are further used for high-level vehicle feature prediction. The details of the producer of this feature pyramid fusion are described in Appendix B.

B. Ground Truth for Four High-Level Vehicle Feature Prediction Subtasks

As displayed in Fig. 1, the multitask learning module has four subtasks for predicting high-level vehicle features

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.

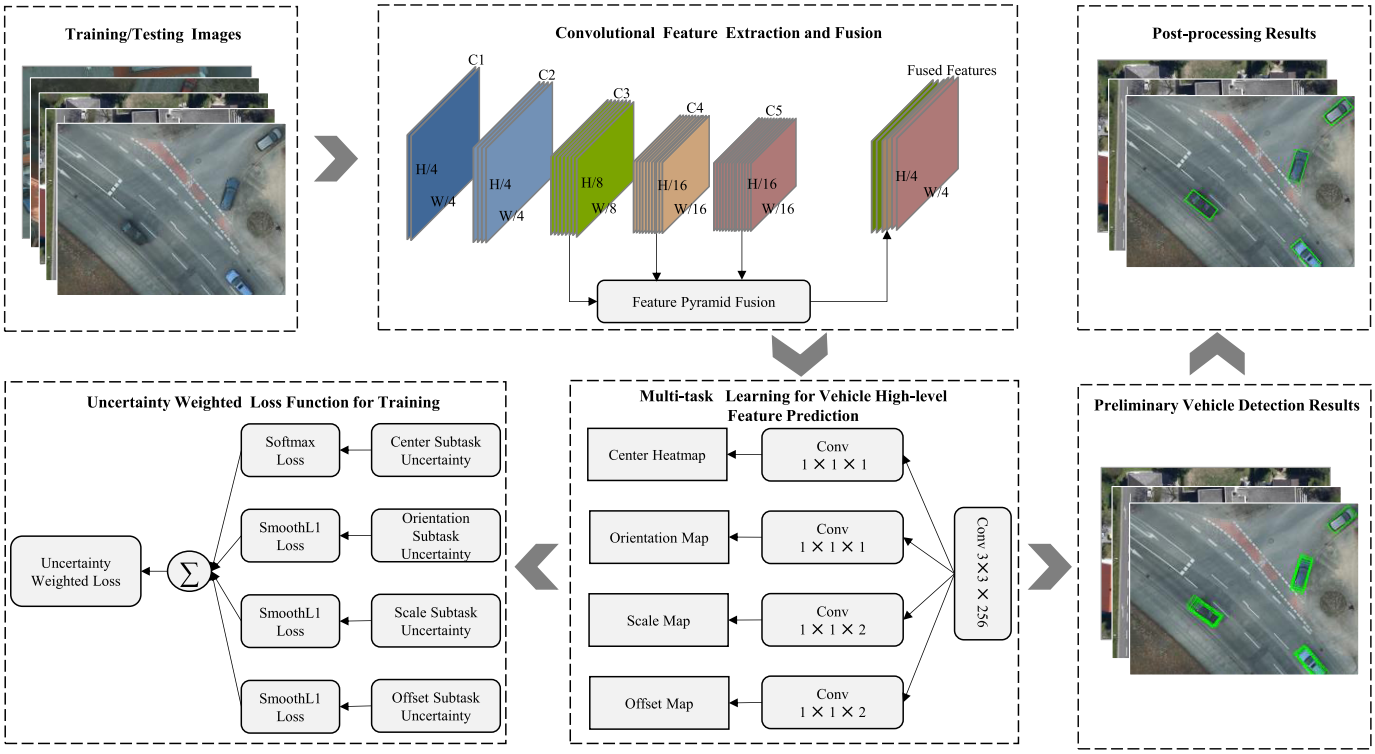


Fig. 1. Overall framework of the proposed oriented vehicle detection approach. Convolutional features generated by different stages of a residual network are combined using a feature pyramid fusion strategy. The multitask learning module consists of four subtasks for vehicle high-level feature prediction. The detection network trained using the uncertainty weighted loss function. For inferencing, the oriented bounding boxes are generated using the predicted vehicle features and a specialized postprocessing method is used to remove redundant results.

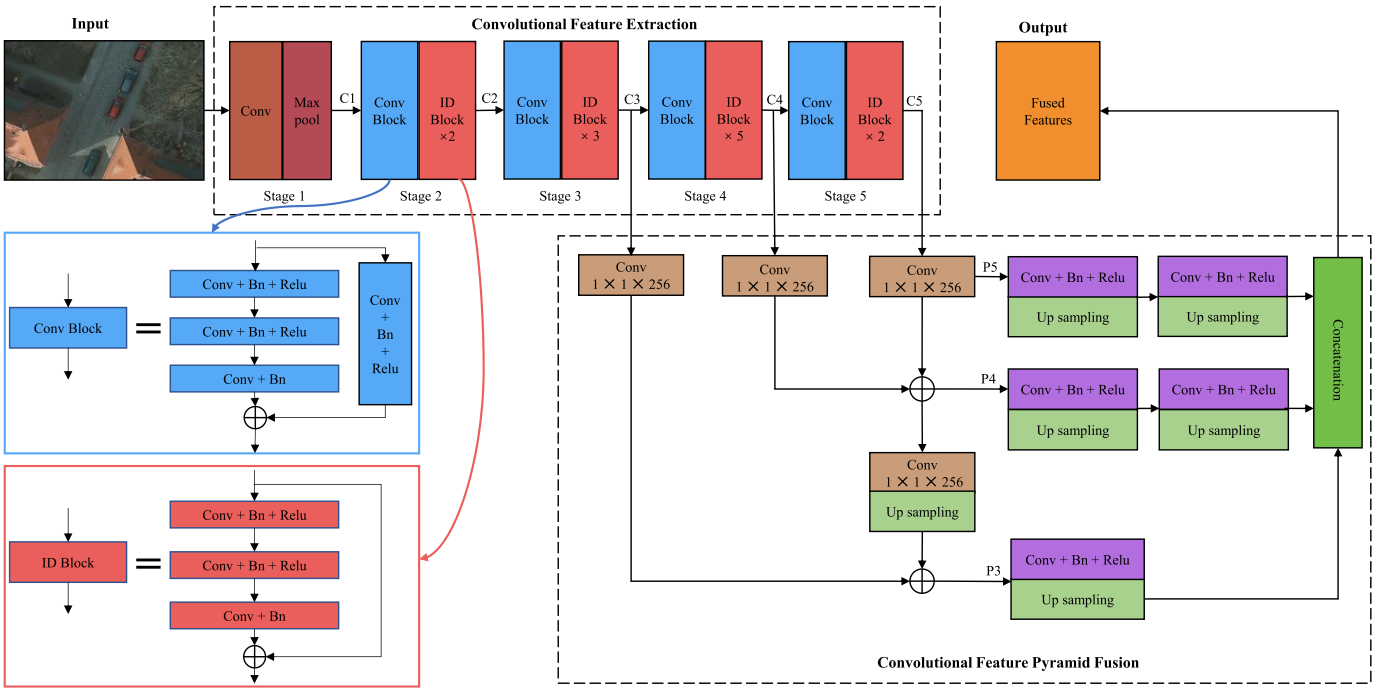


Fig. 2. Convolutional feature extraction and fusion network. Coarse and fine convolutional feature maps (denoted as C1, C2, C3, C4, and C5) are derived from different stages of a residual network. The multiscale feature representation is combined by a feature pyramid fusion strategy. The fusion progress first generates a pyramid of feature maps at different scales (denoted as P3, P4, and P5). Then, the convolutional layers followed by up-sampling operations are applied to up-sample the feature pyramid maps back to the 1/4 of the input image size. Finally, those features with the same resolution are concatenated into comprehensive feature vectors.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.



Fig. 3. Illustration of GT for multitask learning. (a) Oriented bounding box annotations of the vehicle in an aerial image. (b) Vehicle center points, scales, and orientations defined in the proposed approach.

including vehicle centers, scales, orientations, and offsets of the vehicle center points, respectively. Among the four subtasks, vehicle center prediction is a classification subtask and the three others are regression subtasks. First, a $3 \times 3 \times 256$ convolutional layer is used to reduce the fused feature map channel size to 256. Then, four parallel 1×1 convolutional layers are attached to produce a center heatmap, an orientation map, a scale map, and an offset map, respectively. The four predicted maps have the same resolution of concatenated features.

Ground truth (GT) concerning on high-level vehicle features can be obtained with oriented bounding box annotations. One example for vehicle-oriented bounding box annotations and GT of high-level vehicle features is described in Fig. 3. Unlike the horizontal bounding boxes that are usually represented by the coordinates of the upper left corners and the lower right corners, the oriented bounding boxes are represented using five variables including the coordinate of the vehicle center locations, vehicle widths, heights, and orientations.

As shown in Fig. 3(b), the vehicle width is defined as the side of the oriented bounding box where the horizontal axis first encounters in counterclockwise rotation, and the other side is the vehicle height. The orientation is defined as the angle between the horizontal axis and the width edge of the oriented bounding box of a vehicle. It is worth noting that the range of the orientation is between 0° and 90°. For the vehicle center GT, the locations which the vehicle center points fall in are positive, and all other locations are negative. For the scale and orientation GT, the kth positive location is assigned the logarithm values of scale (h, w) and orientation (θ), respectively, where h and w are the height and the width of the kth vehicle, and θ is the kth vehicle orientation. To reduce the ambiguity of positives, the negative locations within a radius 2 of the positives in scale and orientation GT are also assigned values of $(\log(h), \log(w))$, and $\log(\theta)$, while all the other negative locations are assigned zero values.

Considering that the predicted maps are down-sampled as 1/4 of the input image size, some position accuracy may be lost when the locations in the down-sampled maps are remapped to the input image locations. Hence, the offset prediction task is appended to reduce the down-sampled error. For offset GT, the locations around the vehicle center points within radius 2 are assigned values of $((x/4 - \lfloor x/4 \rfloor), (y/4 - \lfloor y/4 \rfloor))$, and all the other locations are assigned zeros, where x and y are the coordinates of the vehicle center points in the input image.

C. Uncertainty Weighted Multitask Loss Function

The multitask learning loss function is composed of one classification loss for vehicle center prediction and three regression loss for vehicle scale, orientation, and offset prediction.

The center classification loss L_{center} adopts the cross-entropy loss. L_{center} is expressed as follows:

$$L_{\text{center}} = -\frac{1}{K}\sum_{i=1}^{W/4}\sum_{j=1}^{H/4}(y_{ij}\log p_{ij} + (1-y_{ij})\log(1-p_{ij})) \tag{1}$$

where K is the number of vehicles, W and H are the width and the height of the input image, respectively, $p_{ij} \in [0, 1]$ represents the network's predicted probability that indicates whether there is a vehicle's center at location (i, j), and $y_{ij} \in \{0, 1\}$ stands for the GT of the vehicle center points.

The scale regression loss $L_{\text{scale}}(i, j; \theta)$ is formulated via the smooth L1 regression loss function. The specific formula is defined as follows:

$$L_{\text{scale}} = \frac{1}{K}\sum_{k=1}^{K}\text{smooth L1}(p_k, y_k) \tag{2}$$

where p_k and y_k are the network's prediction and the GT of the vehicle scale, respectively. Smooth L1 loss is also adopted to model orientation and offset regression loss, which are recorded as L_{orient} and L_{offset}, respectively.

The uncertainty automatically learned from the training data can capture the relative confidence between classification tasks and regression tasks, and thus it can be used to weight losses in multitask learning problems [11]. More details about uncertainty weighted classification/regression loss are described in Appendix C. To jointly learn these four subtasks, the uncertainty weighted loss function is minimized as follows:

$$L = \frac{1}{\sigma_1^2}L_{\text{center}} + \frac{1}{2\sigma_2^2}L_{\text{orient}} + \frac{1}{2\sigma_3^2}L_{\text{scale}} + \frac{1}{2\sigma_4^2}L_{\text{offset}} + \log\sigma_1\sigma_2\sigma_3\sigma_4 \tag{3}$$

where σ_1, σ_2, σ_3, and σ_4 are the learnable uncertainty scalars to balance the weight of each loss terms. In the experiments, the network is trained to predict $\log\sigma^2$ rather than the variance σ^2 to keep numerical stability and avoid any division by zero.

D. Detection Inference of Arbitrarily Oriented Vehicles

The center confidence scores, scales, orientations, and coordinate offsets are predicted by the trained model. The center classification subtask output ($W/4 \times H/4$ 2nd-order tensor) can be treated as a down-sampled classification map for the center and noncenter of vehicles. The scale regression subtask output ($W/4 \times H/4 \times 2$ 2nd-order tensor) represents the width and the height of vehicles. The orientation regression subtask produces the orientation of vehicles ($W/4 \times H/4$ 2nd-order tensor). The offset regression subtask produces the offsets of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.

the vehicle center coordinates ($W/4 \times H/4 \times 2$ 2nd-order tensor), which can adjust the error of the center location. By combining these predicted high-level vehicle features, the oriented bounding boxes with a classification score can be generated and remapped to the image of original size ($W \times H$). The coordinates $\{(v_k^x, v_k^y)|k \in \{0, 1, 2, 3\}\}$ for four vertexes of the oriented vehicle bounding box at the location of $(4i, 4j)$ with the vehicle center confidence score s in the input image can be formulated as follows:

$$\begin{aligned}
&w' = w \times \cos\theta + h \times \sin\theta,\ h' = w \times \sin\theta + h \times \cos\theta \\
&v_0^x = 4i + 4\varepsilon_i + \frac{4w'}{2} - 4w \times \cos\theta,\ v_0^y = 4j + 4\varepsilon_j + \frac{4h'}{2} \\
&v_1^x = v_0^x - 4h \times \sin\theta,\quad v_1^y = v_0^y - 4h \times \cos\theta \\
&v_2^x = v_1^x + 4w \times \cos\theta,\quad v_2^y = v_1^y - 4w \times \sin\theta \\
&v_3^x = v_2^x + 4h \times \sin\theta,\quad v_3^y = v_2^y + 4h \times \cos\theta
\end{aligned} \tag{4}$$

where $w, h, \theta, \varepsilon_i,$ and ε_j, respectively, are the vehicle scales including width, height, vehicle orientation, and vehicle offsets at (i, j).

To filter the noncenter region, only the center locations with relatively high confidence score are selected. However, there are still densely overlapped oriented bounding boxes for a vehicle. The traditional postprocessing method named NMS is usually used to reduce such redundant results. However, the traditional NMS method faces a problem that the IoU computed on the horizontal bounding box is not appropriate when the vehicles are densely arranged. Under this circumstance, the NMS method (i.e., oriented NMS) is extended through computing the IoU on the oriented bounding boxes rather than the horizontal bounding boxes. Considering the fact that the vehicles are usually rectangular and have no overlap with each other, the oriented NMS can be achieved by the following three steps.

1) *Step 1*: Select the oriented box with the highest score in all untreated boxes.
2) *Step 2*: Remove the remaining boxes that have an overlap with the selected box in step 1.
3) *Step 3*: Repeat the above steps until there are no untreated boxes.

IV. Experimental Results and Analysis

In this section, two public aerial image data sets and evaluation metrics are first introduced. Then, the contribution of the vehicle orientation prediction component and the uncertainty weighted loss function is analyzed. Next, the superiority of the proposed vehicle detector is discussed in comparison with the state-of-the-art methods. Finally, some vehicle detection results are displayed to qualitatively illustrate the effectiveness of the proposed approach.

A. Data Set and Training Details

The performance of the proposed vehicle detection approach is demonstrated on two public data sets (i.e., DLR Munich vehicle data set [24] and VEDAI data set [41]). The vehicles in these two data sets are annotated by the oriented bounding boxes. DLR Munich vehicle data set contains 20 aerial images sized 5616×3744 pixels with a spatial resolution of approximately 13 cm. In the experiments, the first ten images in the DLR Munich vehicle data set are used for the training model. These images are further divided into 760×480 subimages with an overlap ratio of 0.5. As a result, a new training data set is reconstructed that contains 2880 subimages and their corresponding annotation files. The rest ten images are also divided into the same size as the training images but without overlapping and then used for testing. Table I describe the details about this data set used for training and testing. The VEDAI data set comprises 1210 RGB tiles sized 1024×1024 pixels at a 12.5-cm spatial resolution. The images are divided into tenfold for cross-validation. Each fold contains approximately the same number of vehicles. In the experiments, all the images in each fold are further divided into 760×480 subimages with an overlap ratio of 0.8. Details regarding for this data set are provided in Table II.

TABLE I
DLR Munich Vehicle Data Set

Item	Training set	Testing set
Image number (Original / Augment)	10 / 2880	10 / 72
Vehicle count (Original / Augment)	3418 / 14612	5892 / 5892

TABLE II
VEDAI Data Set

Item	Training set	Testing set
Image number (Original / Augment)	1089 / 6534	121 / 726
Vehicle count (Original / Augment)	2149 / 3088	237 / 332

The experiments in this article are conducted using Keras. The networks are trained over 10k iterations with a batch size of 4 on a single NVIDIA GTX 1080Ti GPU. The initial learning rate is set as 1×10^{-4} and Adam is used to optimize the network.

B. Evaluation Metrics

To quantitatively evaluate the performance of the proposed approach, four widely used evaluation criteria are adopted, namely, precision, recall, F1 score, and average precision (AP). Those evaluation metrics are defined as follows:

$$\begin{aligned}
\text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FN}} \\
\text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FP}} \\
\text{F1} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}
\end{aligned} \tag{5}$$

where TP is the true positive, FP is the false positive, and FN is the false negative. Generally, a detection result is regarded as a TP if the IoU overlap ratio between the detected bounding box and the GT bounding box is greater than 0.5; otherwise, it is considered as an FP. Moreover, if several detected bounding boxes overlap with the same GT, only one box with the highest overlap ratio is regarded as a TP, and the others are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.

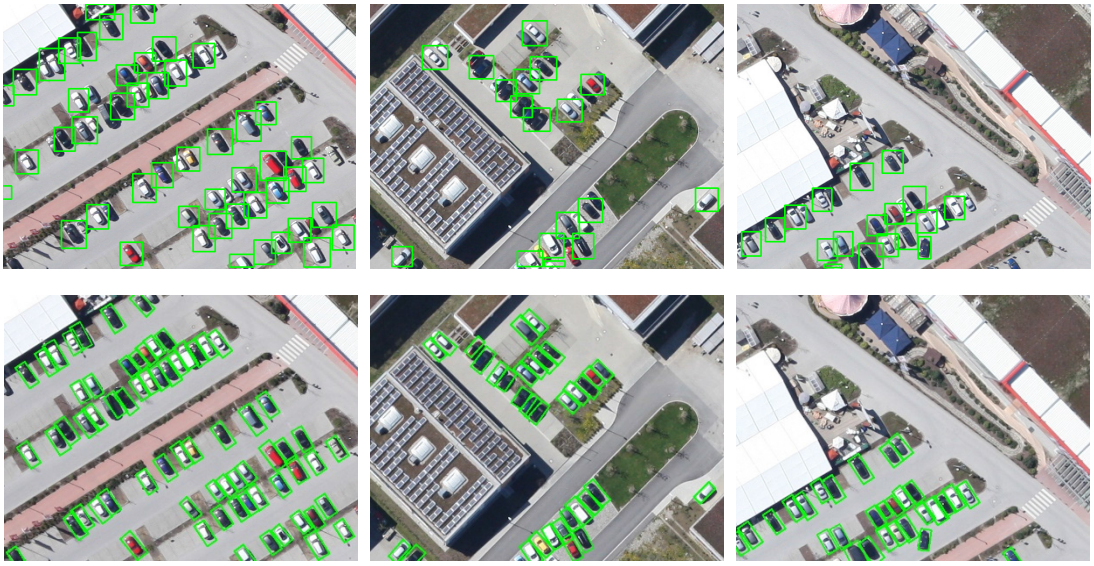


Fig. 4. Qualitative examples showing the vehicle detection results (Top row) without or (Bottom row) with the orientation prediction component. It can be seen that the orientation prediction component manages to achieve higher accuracy of the vehicle detection results in images containing densely parked vehicles.

TABLE III
NETWORK WITH ORIENTATION PREDICTION IMPROVES VEHICLE DETECTION PERFORMANCE

Model	Recall	Precision	F1 score	ΔF1
w/o orientation prediction	80.03%	92.12%	85.7%	—
w/ orientation prediction	86.00%	94.62%	90.1%	+4.4%

FP. TP+FN denotes the number of actual vehicles and is commonly referred as GT, and TP+FP represents the number of vehicle detection results.

The precision and recall metrics measure the ratio of correctly identified vehicles to totally detected vehicles and to actual vehicles, respectively. The comprehensive performance of precision and recall is evaluated using the F1 score.

The AP metric is measured by the area under the precision–recall curve. The higher the F1 score and AP, the better the performance.

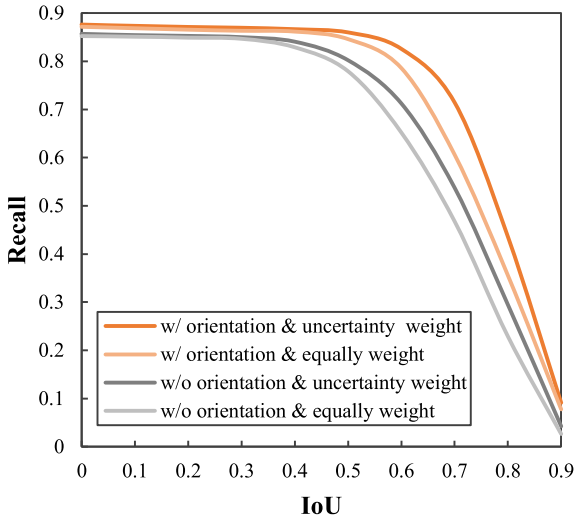


Fig. 5. Recall–IoU curves for different models. The curves show that the recalls generated from the network trained using the uncertainty weighted loss are higher than the equally weighted loss as IoU increases. The curves also display that recalls generated from the network with the orientation prediction module are much higher than without it for any IoU.

C. Vehicle Orientation Prediction Analysis

The prediction of vehicle orientation is a key component of the VCSOP detector. To understand its contribution to vehicle detection performance, a network without the orientation prediction component based on the proposed framework is trained.

Table III shows that adding the orientation prediction component improves vehicle detection performance: 5.97% on recall, 2.5% on precision, and 4.4% on F1 score. Fig. 4 shows some qualitative examples with and without orientation prediction. The images in the first row and the second row, respectively, show the results detected by the network without or with the orientation prediction component. As shown in Fig. 4, the orientation prediction component is particularly helpful for scenes with dense vehicles. This is because dense vehicles have large IoU for their horizontal bounding boxes and are easy to be abandoned after postprocessing.

Fig. 5 displays the recall–IoU trend of four different models, which are trained using the uncertainty or equally weighted loss function for the networks with or without the orientation prediction component. A higher IoU denotes more accurate vehicle detection results. As displayed in Fig. 5, the recalls generated from the network with the orientation prediction component are much higher than those generated from the network without the orientation prediction component in any

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
NETWORKS TRAINED USING UNCERTAINTY WEIGHTED LOSS FUNCTION IMPROVE VEHICLE DETECTION PERFORMANCE

Orientation prediction	Loss Function	Recall	Precision	F1 score	ΔF1
w/o orientation prediction	Equally weighted	77.92%	90.07%	83.6%	—
	Uncertainty weighted	80.03%	92.12%	85.7%	+2.1%
w/ orientation prediction	Equally weighted	84.58%	93.19%	88.7%	—
	Uncertainty weighted	86.00%	94.62%	90.1%	+1.4%

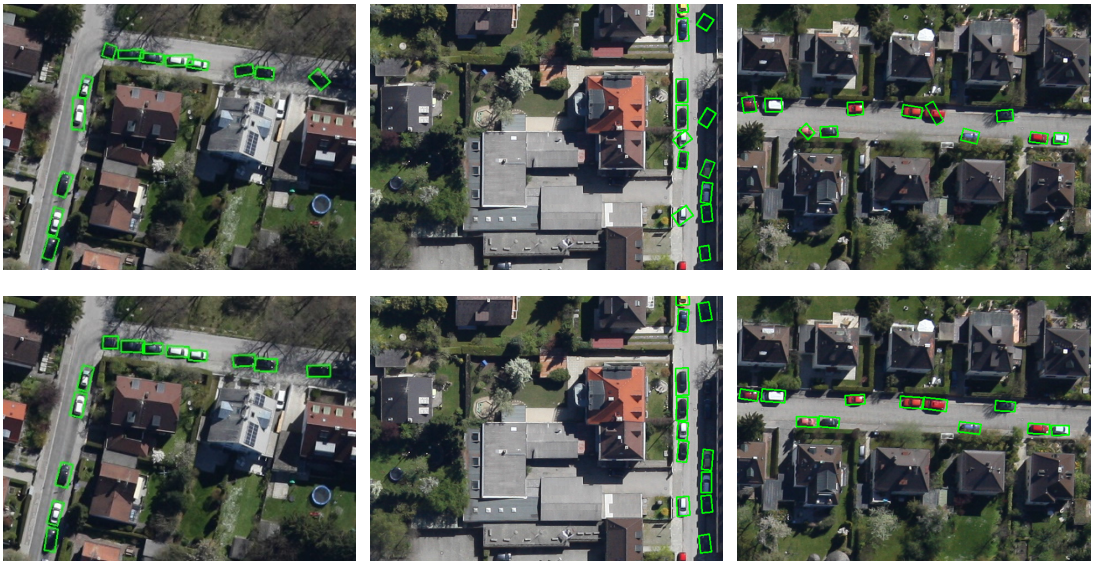


Fig. 6. Qualitative examples showing uncertainty weighted loss helps detect vehicles more accurately (Top row) Vehicle detection results of the network trained using the equally weighted loss. (Bottom row) Detection results of the uncertainty weighted loss.

TABLE V
PERFORMANCE COMPARISON BETWEEN THE PROPOSED VCSOP DETECTOR AND PREVIOUS DETECTION METHODS FOR THE DLR MUNICH VEHICLE DATA SET. THE BEST PERFORMANCES ARE HIGHLIGHTED IN BOLD

Vehicle detection methods	Recall	Precision	F1 score	Time /per image
ACF + soft-cascade AdaBoost [40]	69.30%	86.80%	0.77	4.40 s
Faster R-CNN [16]	68.74%	88.95%	0.78	3.84s
Oriented SSD [49]	78.84%	85.53%	0.82	5.17s
HRPN + Cascade Classifier [48]	78.30%	89.20%	0.83	3.93s
RRPN [42]	82.58%	85.88%	0.84	4.11s
VCSOP detector	**86.00%**	**94.62%**	**0.90**	**2.82s**

IoU. It demonstrates that the orientation task is effective in improving the TPs of the vehicle detection results.

D. Uncertainty Weighted Loss Function Analysis

To demonstrate the effectiveness of the uncertainty weighted loss function, a comparative evaluation of the uncertainty weighted loss and the equally weighted loss function is performed. As presented in Table IV, with the help of the uncertainty weighted loss function, the F1 scores are improved by 2.1% and 1.4%, respectively, for the network without the orientation prediction component (from 83.6% to 85.7%) and the network with the orientation prediction component (from 88.7% to 90.1%).

Fig. 6 shows the vehicle detection results for the network with the orientation prediction component trained using the uncertainty weighted and equally weighted loss. The images in the first row represent the vehicle detection results of the network trained using the equally weighted loss, and the images in the second row show the detection results of the uncertainty weighted loss. As displayed in Fig. 6, the equally weighted loss sometimes fails to predict

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

PERFORMANCE COMPARISON BETWEEN THE PROPOSED VCSOP DETECTOR AND PREVIOUS DETECTION METHODS FOR VEDAI DATA SET. THE BEST MAP IS HIGHLIGHTED IN BOLD

Methods	SVM+LBP [41]	SVM+HOG31 [41]	SVM+HOG31+LBP [41]	RRPN [42]	VCSOP detector
mAP	64.3	71.5	75.0	81.2	**88.5**
Std. Deviation	3.2	2.5	2.2	1.8	**1.4**



Fig. 7. Vehicle detection results of the proposed approach on the DLR Munich vehicle data set. Green, red, and yellow boxes denote the correct, false, and missing vehicle detection results, respectively. (First row to the third row) Dense vehicle detection results from different scenes in urban areas (e.g., vehicles on residential areas, roads, and parking lots). (Last row) Some erroneous or missing vehicle detection results.

accurate vehicle orientations and scales. The uncertainty weighted loss can successfully predict vehicle orientations and scales and produce more accurate vehicle detection results.

Fig. 5 shows that the network trained using the uncertainty weighted loss achieves better recall than the network trained using the equally weighted loss along with the increase in IoU. Take the networks with orientation as an example, the uncertainty weighted loss and the equally weighted loss achieve almost the same recall when the recall is lower than 0.4, but the uncertainty weighted loss achieves better recall when the recall is higher than 0.4. It further demonstrates that the network trained using the uncertainty loss function predicts more accurate vehicle detection results.

E. Comparisons With the State-of-the-Art Methods

In this section, the performance comparisons between the proposed detector and the state-of-the-art methods on two public aerial image data sets are discussed.

1) DLR Munich Vehicle Data Set: Five previous detection methods are compared with the VCSOP detector on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.



Fig. 8. Vehicle detection results of the proposed approach on the VEDAI data set. Green, red, and yellow boxes denote the correct, false, and missing vehicle detection results, respectively. (First row to the third row) Sparse vehicle detection results from different scenes in suburban areas (e.g., vehicles on roads, bare lands, and forest). (Last row) Some erroneous or missing vehicle detection results.

this data set, namely, aggregated channel feature (ACF) + soft-cascade AdaBoost [24], faster R-CNN [16], oriented SSD [49], HRPN + Cascade Classifier [48], and rotation region proposal networks (RRPNs) [42]. The first baseline is based on ACFs [54] and AdaBoost classifier in a soft-cascade structure. Faster R-CNN is one of the state-of-the-art two stage detection methods based on anchor boxes. HRPN + Cascade Classifier is specially designed for vehicle detection in aerial images using hierarchical deep convolutional networks. Oriented SSD is a dedicated model to detect oriented vehicles based on a one-stage detector named SSD. RRPN is an orientation-aware detector using a design of rotated anchor boxes.

Table V reports the performance obtained by the VCSOP detector and the compared methods. As displayed in Table V, VCSOP outperforms the aforementioned methods in terms of recall, precision, and F1 score. Compared with the traditional detector based on handcrafted features, the CNN-based detectors achieve better performance, demonstrating that the deep features can distinguish the difference between vehicles and complicated backgrounds more effectively. Orientation is estimated in oriented SSD to detect vehicles more accurately, and hierarchical feature maps are combined in HRPN + Cascade Classifier to detect small-size objects more effectively. Thus, these methods perform better than faster R-CNN. RRPN obtains satisfactory performance by designing inclined proposals with orientation angle information. By applying the feature pyramid fusion strategy and the orientation prediction module, the proposed method achieved superior performance over the compared methods, which improves the F1 score of 6.0% than the second-best method. The recall and the precision of the VCSOP detector are 3.42% and 8.74% higher than RRPN, respectively, and it suggests that the proposed method can predict accurate detection results and reduce the detection errors.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.

To demonstrate the computational efficiency, the processing speeds of these methods are recorded in Table V. The test time of the VCSOP detector per large-scale aerial image in the DLR Munich vehicle data set is 2.82 s, which is faster than other methods due to directly predicting the vehicle centers and shapes and avoiding the region proposal process.

2) VEDAI Data Set: The images in the VEDAI data set are collected in the suburban areas. Although vehicles in the VEDAI data set are not as dense as in the DLR Munich vehicle data set, this data set is also challenging due to the existence of many disturbance factors (e.g., the shadow of trees and buildings, as well as similar vehicle-like objects). For this data set, the cross-validation process is repeated ten times. In each round, the proposed approach is trained on 6534 images and then tested on 726 images which contain about 332 small land vehicles (slv). The statistic results including average performance and standard deviation over the multiple rounds are reported in Table VI.

The VCSOP detector is compared with three traditional methods and one orientation-aware CNN based method. As presented in Table VI, the comparison of the traditional methods shows that it is an effective way to improve the detection accuracy by combination of different handcraft features. The CNN-based detectors also achieve better performance compared with the traditional methods. The VCSOP detector obtains the best mean AP (mAP), which is 7.3% higher than RRPN. The processing time of the VCSOP detector per large-scale aerial image in this data set is 0.11 s, which is much faster than RRPN. The higher accuracy and faster speed in this data set further demonstrate the effectiveness and efficiency of the proposed method.

F. Qualitative Results

To qualitatively illustrate the effectiveness of the proposed approach, some vehicle detection results from different scenes are displayed in Fig. 7 for the DLR Munich vehicle data set and in Fig. 8 for the VEDAI data set. In these images, green boxes represent the correct vehicle detection results, red boxes represent the false results, and yellow boxes represent the missing results.

As displayed in Figs. 7 and 8, the VCSOP detector can obtain a good performance in spite of complex scenes and large variations in vehicles. The images from the first row to the third row in Fig. 7 display the dense vehicle detection results for different scenes in urban areas (e.g., vehicles on residential areas, roads, and parking lots). The first three rows in Fig. 8 show sparse vehicle detection results from the scenes in suburban areas such as bare lands and forest. These images show that the vehicles can be identified accurately despite that the scenes are complex and the vehicles are partly occluded by the shadow of trees and buildings. However, there are still some false detections and missing detection results, as displayed in the last row in Figs. 7 and 8. The false detections are mainly caused by similar objects like road marks, air conditioning units on buildings, and wretched boats. The missing detections mostly occur once the vehicles are seriously occluded by shadow of trees and buildings or vehicle locations near image block boundaries.

V. Conclusion

Vehicle detection is an important and challenging task in the field of remote sensing. Most existing methods detect horizontal bounding boxes of vehicles using traditional detectors based on sliding windows or CNN detectors based on anchor boxes. The proposed VCSOP detector is a one-stage, anchor-free, and orientation-aware vehicle detection method. The procedure is composed of four subtasks that can directly predict high-level vehicle features via a fully convolutional network, that is, one classification subtask to identify vehicle central points, and three regression subtasks to predict vehicle orientations, scales, and offsets of the vehicle central points. Besides, the weight of the multitask loss function can be automatically learned from the training data according to the uncertainty of each subtask. For inference, an oriented NMS postprocessing that computes the IoU on the oriented bounding box is adopted to reduce redundant results. The VCSOP detector was verified on two large high-resolution aerial image data sets, namely, the DLR Munich vehicle data set and the VEDAI data set. Both the quantitative and qualitative results of the experiments demonstrate the effectiveness of the VCSOP detector. In future work, we will consider combining multiple object tracking methods with the proposed approach to derive dynamic vehicle trajectories.

Appendix

A. Details of the Residual Network to Extract Coarse and Fine Convolutional Features

The coarse and fine convolutional features are extracted from a ResNet-50 [52], which adds shortcut connection in plain network to make the gradient directly backpropagate to earlier layers. ResNet-50 has two main types of residual blocks, namely, identity block and convolutional block. The identity block directly connect the layers with the same input and output dimensions. The convolutional block adds a 1×1 convolution layer in the shortcut path to match up the input and output dimensions. According to the residual network structure, its convolutional layers can be divided into five stages. In the first stage, a 7×7 convolutional layer with a stride of 2 followed by a max pooling layer is applied to generate a feature map as 1/4 size of the input image size. The last four stages are composed of 3, 4, 6, and 3 residual blocks and generate convolutional feature maps as 1/4, 1/8, 1/16, and 1/32 of the input size, respectively. Specifically, to better localize small-sized vehicles, the down-sampling layer is removed in stage 5 and a dilated convolution layer is used to maintain the final feature map as 1/16 of the input size, which is similar to [53].

B. Details of the Feature Pyramid Fusion Strategy

As displayed in Fig. 2, C3, C4, and C5 outputted from the residual network are combined via this feature pyramid fusion strategy. The feature fusion starts from the deepest feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.

maps C5; first, a 1×1 convolutional layer is applied to C5 to reduce its feature dimensions into 256 (denoted as P5), and then P5 is merged with C4 via element-wise feature addition, where C4 also undergoes a 1×1 convolutional layer to adjust its feature dimensions into the same number. The merged features are denoted as P4. Next, P4 is up-sampled by a factor of 2, and then the unsampled feature is added to C3, which is connected to a $1 \times 1 \times 256$ convolutional layer. This added feature is denoted as P3. In this way, a pyramid of feature maps (P3, P4, and P5) is constructed. To get the desired resolution, P3, P4, and P5 are, respectively, performed two, two, and one up-sampling module to back their spatial resolution into 1/4 scale, where the up-sampling module is composed of a 3×3 convolutional layer, batch normalization, ReLU, and an up-sampling layer. Finally, the up-sampled features with the same resolution are fused together using a concatenation operation.

C. Uncertainty Weighted Classification and Regression Loss

The multitask learning loss function can be defined as follows:

$$L(i) = \sum_{i=1}^{N} w_i L_i(\theta) \tag{6}$$

where N is the number of subtasks, L_i is the corresponding subtask loss function to be minimized with the network parameters θ, and each subtask loss L_i is weighted by a scalar w_i. In [11], the uncertainty automatically learned from the training data is used to weight losses in multitask learning problems.

For a classification subtask, the probabilistic outputs are squashed through a softmax function with an observation noise scalar σ

$$p(y \mid f(\theta), \sigma) = \text{softmax}\left(\frac{1}{\sigma^2} f(\theta)\right) \tag{7}$$

where $f(\theta)$ is the output of a neural network with parameter θ. An assumption in [11] is formulated as follows:

$$\frac{1}{\sigma^2} \sum_{c'} \exp\left(\frac{1}{\sigma^2} f_{c'}(\theta)\right) \approx \left(\sum_{c'} \exp(f_{c'}(\theta))\right)^{\frac{1}{\sigma^2}}. \tag{8}$$

By applying it, the negative log likelihood (NLL) loss for classification using the scaled softmax likelihood can be simplified as

$$L_{\text{class}}(\theta, \sigma) = \frac{1}{\sigma^2} L_{\text{class}}(\theta) + \log \sigma \tag{9}$$

where $L_{\text{class}}(\theta)$ denotes the cross-entropy loss for classification without the noise scalar σ.

For a regression subtask, the likelihood is defined as a normal distribution

$$p(y \mid f(\theta), \sigma) = \mathcal{N}(f(\theta), \sigma^2). \tag{10}$$

Then, the NLL loss for regression can be written as

$$L_{\text{regress}}(\theta, \sigma) = \frac{1}{2\sigma^2} L_{\text{regress}}(\theta) + \log \sigma \tag{11}$$

where $L_{\text{regress}}(\theta)$ is a Euclidean loss for regression without the noise scalar σ.

In multitask outputs, the joint likelihood factorizes over subtasks given sufficient statistics, that is

$$p(y_1, \ldots, y_k|\theta, \sigma) = p(y_1|\theta, \sigma) \ldots p(y_k|\theta, \sigma). \tag{12}$$

Thus, the NLL loss for multitask learning problem including a classification and a regression subtask can be derived as follows:

$$L(\theta, \sigma_1, \sigma_2) = \frac{1}{\sigma_1^2} L_{\text{class}}(\theta) + \frac{1}{2\sigma_2^2} L_{\text{regress}}(\theta) + \log \sigma_1 \sigma_2 \tag{13}$$

where σ_1 and σ_2 represent the uncertainty scalars, which are automatically learned from the training data to weight the multitask loss.

REFERENCES

[1] Y. Wang, D. Zhang, Y. Liu, B. Dai, and L. H. Lee, "Enhancing transportation systems via deep learning: A survey," *Transp. Res. C, Emerg. Technol.*, vol. 99, pp. 144–163, Feb. 2019.

[2] H. Zhou, D. Creighton, L. Wei, D. Y. Gao, and S. Nahavandi, "Video driven traffic modelling," in *Proc. IEEE/ASME Int. Conf. Adv. Intell. Mechatronics*, Jul. 2013, pp. 506–511.

[3] W. Liu, F. Yamazaki, and T. T. Vu, "Automated vehicle extraction and speed determination from QuickBird satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 4, no. 1, pp. 75–82, Mar. 2011.

[4] G. Kopsiaftis and K. Karantzalos, "Vehicle detection and traffic density monitoring from very high resolution satellite video data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2015, pp. 1881–1884.

[5] X. Cao, C. Wu, J. Lan, P. Yan, and X. Li, "Vehicle detection and motion analysis in low-altitude airborne video under urban environment," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 21, no. 10, pp. 1522–1533, Oct. 2011.

[6] S. Hinz, "Detection and counting of cars in aerial images," in *Proc. Int. Conf. Image Process.*, Sep. 2003, pp. 997–1000.

[7] A. Voulodimos, N. Doulamis, A. Doulamis, and E. Protopapadakis, "Deep learning for computer vision: A brief review," *Comput. Intell. Neurosci.*, vol. 2018, pp. 1–13, Feb. 2018.

[8] J. Huang *et al.*, "Speed/accuracy trade-offs for modern convolutional object detectors," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3296–3297.

[9] S. M. M. Kahaki, M. D. J. Nordin, and A. Hossein Ashtari, "Incident detection algorithm based on radon transform using high-resolution remote sensing imagery," in *Proc. Int. Conf. Electr. Eng. Informat.*, Jul. 2011, pp. 1–5.

[10] R. Pelapur, K. Palaniappan, and G. Seetharaman, "Robust orientation and appearance adaptation for wide-area large format video object tracking," in *Proc. IEEE 9th Int. Conf. Adv. Video Signal-Based Surveill.*, Sep. 2012, pp. 337–342.

[11] R. Cipolla, Y. Gal, and A. Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7482–7491.

[12] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[13] J. R. R. Uijlings, K. E. A. van de Sande, T. Gevers, and A. W. M. Smeulders, "Selective search for object recognition," *Int. J. Comput. Vis.*, vol. 104, no. 2, pp. 154–171, Apr. 2013.

[14] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.

[15] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[16] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[17] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[18] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.

[19] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7263–7271.
[20] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2016, pp. 21–37.
[21] T. Kong, F. Sun, A. Yao, H. Liu, M. Lu, and Y. Chen, "RON: Reverse connection with objectness prior networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5936–5944.
[22] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, Feb. 2020.
[23] Y. Yuan, Z. Xiong, and Q. Wang, "VSSA-NET: Vertical spatial sequence attention network for traffic sign detection," *IEEE Trans. Image Process.*, vol. 28, no. 7, pp. 3423–3434, Jul. 2019.
[24] H. Law and J. Deng, "Cornernet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 734–750.
[25] A. Newell, Z. Huang, and J. Deng, "Associative embedding: End-to-end learning for joint detection and grouping," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2017, pp. 2277–2287.
[26] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.
[27] X. Zhou, J. Zhuo, and P. Krahenbuhl, "Bottom-up object detection by grouping extreme and center points," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 850–859.
[28] W. Liu, S. Liao, W. Ren, W. Hu, and Y. Yu, "High-level semantic feature detection: A new perspective for pedestrian detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5187–5196.
[29] T. Zhao and R. Nevatia, "Car detection in low resolution aerial image," in *Proc. 8th IEEE Int. Conf. Comput. Vis. ICCV*, Jul. 2001, pp. 710–717.
[30] T. Moranduzzo and F. Melgani, "A SIFT-SVM method for detecting cars in UAV images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 6868–6871.
[31] T. Moranduzzo and F. Melgani, "Automatic car counting method for unmanned aerial vehicle images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 3, pp. 1635–1647, Mar. 2014.
[32] T. Moranduzzo and F. Melgani, "Detecting cars in UAV images with a catalog-based approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 10, pp. 6356–6367, Oct. 2014.
[33] Z. Chen *et al.*, "Vehicle detection in high-resolution aerial images via sparse representation and superpixels," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 103–116, Jan. 2016.
[34] H. Zhou, L. Wei, C. P. Lim, D. Creighton, and S. Nahavandi, "Robust vehicle detection in aerial images using bag-of-words and orientation aware scanning," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7074–7085, Dec. 2018.
[35] W. Shao, W. Yang, G. Liu, and J. Liu, "Car detection from high-resolution aerial imagery using multiple features," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 4379–4382.
[36] H. Grabner, T. T. Nguyen, B. Gruber, and H. Bischof, "On-line boosting-based car detection from aerial images," *ISPRS J. Photogramm. Remote Sens.*, vol. 63, no. 3, pp. 382–396, May 2008.
[37] A. Kembhavi, D. Harwood, and L. S. Davis, "Vehicle detection using partial least squares," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 6, pp. 1250–1265, Jun. 2011.
[38] Q. Tan, J. Wang, and D. A. Aldred, "Road vehicle detection and classification from very-high-resolution color digital orthoimagery based on object-oriented method," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Jul. 2008, pp. 475–478.
[39] J. Leitloff, D. Rosenbaum, F. Kurz, O. Meynberg, and P. Reinartz, "An operational system for estimating road traffic information from aerial images," *Remote Sens.*, vol. 6, no. 11, pp. 11315–11341, Nov. 2014.
[40] K. Liu and G. Mattyus, "Fast multiclass vehicle detection on aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 9, pp. 1938–1942, Sep. 2015.
[41] S. Razakarivony and F. Jurie, "Vehicle detection in aerial imagery: A small target detection benchmark," *J. Vis. Commun. Image Represent.*, vol. 34, pp. 187–203, Jan. 2016.
[42] J. Ma *et al.*, "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Trans. Multimedia*, vol. 20, no. 11, pp. 3111–3122, Nov. 2018.
[43] X. Chen, S. Xiang, C.-L. Liu, and C.-H. Pan, "Vehicle detection in satellite images by hybrid deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 10, pp. 1797–1801, Oct. 2014.
[44] T. Qu, Q. Zhang, and S. Sun, "Vehicle detection from high-resolution aerial images using spatial pyramid pooling-based deep convolutional neural networks," *Multimedia Tools Appl.*, vol. 76, no. 20, pp. 21651–21663, Oct. 2017.
[45] N. Ammour, H. Alhichri, Y. Bazi, B. Benjdira, N. Alajlan, and M. Zuair, "Deep learning approach for car detection in UAV imagery," *Remote Sens.*, vol. 9, no. 4, p. 312, Mar. 2017.
[46] T. Tang, S. Zhou, Z. Deng, L. Lei, and H. Zou, "Fast multidirectional vehicle detection on aerial images using region based convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2017, pp. 1844–1847.
[47] Z. Deng, H. Sun, S. Zhou, J. Zhao, and H. Zou, "Toward fast and accurate vehicle detection in aerial images using coupled region-based convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3652–3664, Aug. 2017.
[48] T. Tang, S. Zhou, Z. Deng, H. Zou, and L. Lei, "Vehicle detection in aerial images based on region convolutional neural networks and hard negative example mining," *Sensors*, vol. 17, no. 2, pp. 3361–33617, Feb. 2017.
[49] T. Tang, S. Zhou, Z. Deng, L. Lei, and H. Zou, "Arbitrary-oriented vehicle detection in aerial imagery with single convolutional neural networks," *Remote Sens.*, vol. 9, no. 11, p. 1170, Nov. 2017.
[50] K. Zheng, M. Wei, G. Sun, B. Anas, and Y. Li, "Using vehicle synthesis generative adversarial networks to improve vehicle detection in remote sensing images," *ISPRS Int. Geo-Inf.*, vol. 8, no. 9, p. 390. Sep. 2019.
[51] Q. Wang, S. Liu, J. Chanussot, and X. Li, "Scene classification with recurrent attention of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1155–1167, Feb. 2019.
[52] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[53] T. Song, L. Sun, D. Xie, H. Sun, and S. Pu, "Small-scale pedestrian detection based on topological line localization and temporal feature aggregation," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 536–551.
[54] P. Dollar, R. Appel, S. Belongie, and P. Perona, "Fast feature pyramids for object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 8, pp. 1532–1545, Aug. 2014.



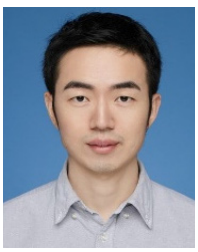
Furong Shi received the B.S. degree from the China University of Petroleum, Qingdao, China, in 2017. She is pursuing the Ph.D. degree with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China.

Her research interests include machine learning and remote sensing image processing.



Tong Zhang received the M.Eng. degree in cartography and geographic information system (GIS) from Wuhan University, Wuhan, China, in 2003, and the Ph.D. degree in geography from San Diego State University, San Diego, CA, USA, and the University of California at Santa Barbara, Santa Barbara, CA, in 2007.

He is a Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. His research topics include high-resolution remote sensing image analysis, computer vision, urban computing, and machine learning.



Tao Zhang received the B.S. degree in surveying and mapping from Central South University, Changsha, China, in 2014, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2019.

He is an Engineer with the Tianjin Institute of Geotechnical Investigation Surveying, Tianjin, China. His research interests include high-resolution image processing and urban remote sensing applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:48:51 UTC from IEEE Xplore. Restrictions apply.